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MyDesign Portfolio

Element A Initial Template

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

Bicycling has become increasingly popular in cities and towns as a way to commute, exercise, and reduce environmental impact. However, unsafe bike routes remain a serious problem that threatens riders' safety and discourages cycling as a regular form of transportation. In the United States, bicyclist deaths have been steadily rising, with most occurring in urban areas with heavy traffic and little bike infrastructure (National Safety Council, 2023). Studies show that protected bike lanes can reduce crashes by up to 90 percent, while painted lanes provide only minimal safety improvements (National Association of City Transportation Officials, 2014). Despite the evidence, many communities lack adequate infrastructure because of cost concerns. Building bike lanes can range from \$5,000 per mile for simple painted lanes to \$600,000 per mile for protected lanes with barriers (PeopleForBikes, 2022). This problem affects entire communities—cyclists, drivers, families, and city planners—while also connecting to broader issues of environmental sustainability and accessibility. The problem is that many bikers don't feel safe when biking on main roads.

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The problem of unsafe bike routes involves cyclists, families, city governments, and drivers, all of whom are affected by the lack of safe infrastructure. The main cause is that many communities either do not have bike lanes at all or rely only on painted lanes, which do little to protect riders from traffic. This issue occurs most often in busy urban areas where cars and bicycles share crowded roads, but it can also be found in suburban towns that never invested in proper bike design. The size of the problem is national, as bicyclist deaths in the United States have risen in recent years, and it extends globally in cities where cycling is growing in popularity without the safety measures to match. Physically, the problem happens on streets and intersections where cyclists must compete with vehicles, especially during commuting hours when traffic is heaviest. It happens year-round in most places, but the risks are higher during warm-weather months when more people ride. The problem continues because governments often delay or avoid investing in costly but safer protected bike lanes. As a result, bike-related crashes and injuries occur regularly, with fatalities averaging more than two cyclists every day in the U.S.

The problem of unsafe bike routes is important because it impacts lives, health, and the environment. Bicyclist deaths continue to rise in urban areas where traffic is dense and bike lanes are limited (National Safety Council, 2023). At the same time, data shows that when protected lanes are built, cycling increases and accident rates decrease significantly



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(NACTO, 2014). Costs remain a major barrier, but research demonstrates that investing in protected bike lanes is far less expensive than the long-term costs of traffic injuries, deaths, and pollution (PeopleForBikes, 2022). Solving this problem now is essential for creating safer communities, encouraging sustainable transportation, and protecting the lives of cyclists across the country.

Unsafe bike routes also connect to other social and environmental challenges. Lack of safe cycling infrastructure leads to more car dependency, which increases air pollution and greenhouse gas emissions. Communities without affordable, safe bike lanes face unequal access to mobility, especially among lower-income groups who rely on bicycles for daily transportation (Streetsblog, 2020). A prototype that helps design safer, more cost-effective bike lanes could benefit not only riders but also entire cities by reducing congestion, lowering emissions, and improving public health.

1. Primary stakeholders in this problem include cyclists, who are most directly at risk of injury or death without safe infrastructure.
2. Families of cyclists are also impacted, as safety concerns discourage children or teenagers from biking to school or activities.
3. Local governments and city planners are stakeholders as well, since they are responsible for designing and funding safe infrastructure.
4. Secondary stakeholders include drivers, who benefit from reduced road conflicts,



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- and communities as a whole, which gain cleaner air and improved public health.
5. Cyclists want physical protection from traffic, families want safety and accessibility, and governments need cost-effective solutions. Addressing these concerns highlights the urgency of solving this problem.

National Association of City Transportation Officials. *Urban Bikeway Design Guide*. Island Press, 2014.

National Safety Council. "Bicycle Deaths." *Injury Facts*, 2023,
<https://injuryfacts.nsc.org/home-and-community/safety-topics/bicycle-deaths/>.

PeopleForBikes. *Protected Bike Lanes Mean Business*. 2022.

Streetsblog USA. "Meet the Protected Bike Lane That Any City Can Afford to Build." 29 July 2020,
<https://usa.streetsblog.org/2020/07/29/meet-the-protected-bike-lane-that-any-city-can-afford-to-build>.



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